

DANIEL MAGRO

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Portfolio Website: magro.dev | Markets, financial infrastructure, and technical systems

PROFESSIONAL SUMMARY

Markets and technology professional with 13 years in institutional trading, now building Ethereum infrastructure, AI agent systems, and security-focused blockchain tooling. Combines market-structure judgment with hands-on development across Go, Solidity, Rust, Python, and TypeScript.

PROFESSIONAL EXPERIENCE

Full-Stack, AI & Web3 Development | Freelance + RAMM.ai | Remote / New York, NY | 03/2024 - Present

- Freelance AI Agent Development & Infrastructure (08/2025 - Present): Build and deploy Hermes-based agentic workflows, multi-agent orchestration systems, and open-source tooling. Develop AI agent observability tools, structured metadata systems for LLM consumption, and agent evaluation frameworks. Active crypto and DeFi trading across Ethereum, Solana, and Layer 2 networks.
- RAMM.ai (03/2024 - 07/2025): Developed EVM marketplace contracts, the product frontend, and public website for an early-stage Web3 startup; translated on-chain mechanics into product requirements and presented the platform to retailers, partners, and users.
- Apply active crypto trading and protocol research to software design, with particular focus on liquidity, AMMs, execution, DeFi mechanics, market infrastructure, and risk.

Medical Leave, Recovery & Technical Retraining | Remote | 08/2022 - 02/2024

- Took medical leave after a severe injury requiring hospitalization and multiple facial and nasal reconstructive surgeries.
- Completed an extended recovery period that also included travel and family caregiving.
- Used the period for intensive computer science, Solidity, blockchain, and AI study through Harvard CS50, Boot.dev, Cyfrin Updraft, independent protocol research, and active crypto and DeFi trading.

Independent Systems Developer & Trader | Self-Employed | Remote | 05/2019 - 07/2022

- Designed Python-based monitoring, execution, and automation workflows across multiple platforms; analyzed secondary-market liquidity, pricing, and operational risk.
- Built automated systems for market scanning, opportunity identification, and execution in constrained opportunity sets. Applied portfolio management principles from institutional trading to independent operations.
- Scaled operations into a full-time business with three consecutive years of six-figure profitability through systematic execution and risk management.

VP, Fixed Income Portfolio Manager | PGIM | Newark, NJ | 04/2017 - 04/2019

- Managed global interest-rate and relative-value portfolios across U.S., Canadian, European, and Japanese markets, achieving a personal Sharpe ratio of 2.1 and contributing to multi-billion-dollar portfolio outperformance.

VP, Asian Hours Macro Execution | PointState Capital | New York, NY | 05/2015 - 02/2017

- Translated portfolio-manager intent into cross-asset fixed income, FX, equity, and derivatives execution as the real-time interface with global trading desks during Asian market hours.

VP, Proprietary Trading | Nomura Securities | New York, NY | 03/2011 - 05/2015

- Traded global rates, FX, and equities across U.S., European, and Japanese markets, delivering positive P/L annually with a personal Sharpe ratio of 2.4 versus a desk average of approximately 1.8.

Institutional Fixed Income Sales | Merrill Lynch & Jefferies | New York, NY | 07/2006 - 02/2011

- Covered institutional interest-rate clients within senior sales pods generating approximately \$45MM in annual production at Merrill Lynch and \$10MM at Jefferies; translated market structure, liquidity, and rates-product mechanics for institutional clients.

SELECTED TECHNICAL WORK

AMM Simulation Engine | magro.dev/projects/eth-amm-sim | Live: eth-amm-sim.vercel.app

Interactive automated market maker lab exploring impermanent loss, fee accrual, concentrated liquidity, and fixed-point math. Built with Go, TypeScript, and Solidity, featuring Monte Carlo simulation and Vite frontend.

Rollup Mechanics Lab | magro.dev/projects/eth-l2 | Live: eth-l2.vercel.app

Comparative Ethereum L2 implementation demonstrating fraud proofs versus validity proofs for handling invalid transaction batches. Go and Solidity with EVM tracing, deployed via Cloudflare Tunnel.

Agent Runtime | magro.dev/projects/agent-runtime | Live: agent-runtime-rho.vercel.app

Interactive architecture visualization showing how tool-using AI agents process requests through session hydration, iterative loops, tool dispatch, verification, and state persistence.

TECHNICAL PROFICIENCY

Languages: Go, Solidity, TypeScript, Python, Rust, C

Frameworks and Tools: Next.js, Foundry, Anchor, Cobra, Vite

Infrastructure: Ethereum, Solana, EVM, JSON-RPC, Beacon API, Cloudflare Tunnel

AI and Agents: Hermes workflows, LLM evaluation, agent observability, llms.txt, structured metadata

Specialized: EVM tracing, MEV detection, fraud proofs, secp256k1, fixed-point math, Monte Carlo simulation

SELECTED WRITING AND RESEARCH

Octopus: Agentic Systems | magro.dev/agent-research/octopus-agentic-systems

AI Infrastructure Financing | magro.dev/agent-research/ai-infrastructure-financing

The Subprime AI Data Center Narrative | magro.dev/writing/subprime-ai-data-center-narrative

EDUCATION AND TECHNICAL TRAINING

B.S. Finance, Pennsylvania State University - Magna Cum Laude | Overall GPA: 3.73 | Major GPA: 3.93

Technical Training: Cyfrin Updraft | Boot.dev | Harvard CS50 | MIT OpenCourseWare | DeepLearning.AI

Community: Local library volunteer assisting new citizens with documentation and English; animal-shelter volunteer.

Interests: Pickleball, softball, and avid reading in the history of finance, economics, military affairs, and civilizations.